

EE 115 Homework 6 Solutions

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1. RL differentiator

$$H_{RL}(j2\pi f) = \frac{j2\pi fL}{R + j2\pi fL}. \quad (1)$$

The spectrum of $u_{FM}(t)$ is confined to $f \in [f_c - W/2, f_c + W/2]$, so the largest frequency component is $f_{\max} = f_c + W/2$. For $2\pi fL \ll R$ the magnitude reduces to

$$|H_{RL}(j2\pi f)| \approx \frac{2\pi fL}{R},$$

which is linear in f and therefore behaves like an ideal differentiator (scaled by L/R) over the signal band. Thus the RL network can replace the differentiator provided its corner frequency is well above the FM band:

$$2\pi(f_c + W/2)L \ll R \quad \text{or equivalently} \quad \frac{R}{L} \gg 2\pi(f_c + W/2).$$

2. RLC differentiator

In the underdamped case $\frac{1}{R^2C^2} \ll \frac{4}{LC}$ the poles are $p_{1,2} = -\sigma \pm j\omega_0$ with $\omega_0 = 1/\sqrt{LC}$, so

$$H_{RLC}(j2\pi f) = \frac{j2\pi f/(RC)}{-4\pi^2 f^2 + j\frac{2\pi f}{RC} + \frac{1}{LC}}.$$

(a) For $f \ll f_0 = \frac{1}{2\pi\sqrt{LC}}$ the denominator is dominated by $\frac{1}{LC}$, so

$$|H_{RLC}(j2\pi f)| \approx \frac{2\pi fL}{R},$$

a straight line with slope $\alpha = \frac{2\pi L}{R}$. Choosing the FM carrier f_c in this low-frequency rising region and keeping the whole band below resonance,

$$f_c + W/2 \ll f_0,$$

makes the curve locally linear. Around such an f_c ,

$$|H_{RLC}(j2\pi f)| \approx H_0 + \alpha(f - f_c), \quad H_0 = |H_{RLC}(j2\pi f_c)| \approx \alpha f_c,$$

i.e., the magnitude behaves as $c + \alpha f$ for $|f - f_c| < W/2$. A sketch would show the magnitude rising almost linearly from 0, bending over, and peaking near f_0 ; the tangent at $f = f_c$ on the rising part is the desired linear approximation.

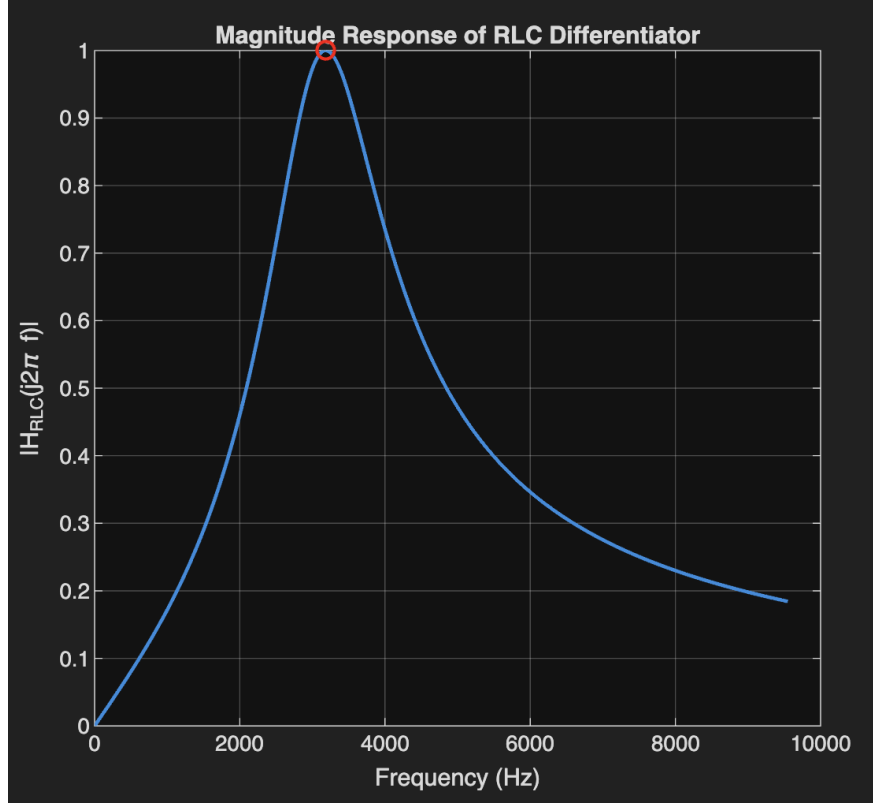


Figure 1: Sketch of $|H_{RLC}(j2\pi f)|$ versus frequency, showing the linear region near f_c below f_0 .

(b) Advantages over the RL circuit:

- Band-pass behavior around f_c (with $f_c < f_0$) suppresses broadband noise away from the FM band; the RL high-pass passes all higher-frequency noise.
- The resonant gain $Q = \frac{1}{R} \sqrt{\frac{L}{C}}$ allows much larger differentiator gain near f_c (and a peak near f_0), whereas the RL magnitude is always ≤ 1 . Hence the RLC gives better SNR and steeper slope for demodulation.